

Video Transcript

Module 1: Introduction to the Artificial Intelligence Design Process

Video 1: Introduction to the AI Design Process (03:01)

In this module, we will learn about four activities required to design an AI product. We will describe each of these four activities and illustrate their importance in the design process by using examples of different AI technologies with a focus on Natural Language Processing. Before we dive in, let me give you two reasons why the design process in AI is so important.

The first reason is because of the rapid evolution of AI possibilities. This means that finding practical applications almost always requires some form of innovation. Indeed, it is very hard to copy an AI application, and they are almost always unique. This is very much unlike in other areas where you can do as your neighbor and practically get the same car, clothes, phone, tennis racket, education, and even house. In AI, copying what others do without some design work will probably result in sub-optimal outcomes.

A second reason, co-founding with the first, is the importance of AI complementary assets. Very often it is not that AI in itself provides value. What does provide value is the combination of AI with changes in business processes. In some cases where you need a piece of AI that does not exist such as a predictive model unique to your company or a specific use of computer vision to distinguish yourself from the competition, one has to design a deployment process where the organization will keep improving the technology by feeding the right training data to the machine learning algorithms and at the same time will be adapting their business processes and products to benefit from such gradual improvements. Thus, one has to design AI products with a vision of how the organization will adapt the technologies deployed and improve.

I suggest that before you continue, you stop for a few minutes and identify some AI products you would like to design. For each of these design ambitions, keep a log throughout the course where you identify how "what" you're learning relates to each of them. By the end of this module, you will be able to organize the design effort into four activities. By the end of the course, the objective is to prepare you for many successful AI product design efforts.